

Institute of Actuaries of India

Subject ST4 – Pension & Other Employee Benefits

March 2017 Examination

INDICATIVE SOLUTION

Introduction:-

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

i)

- Tax concessions can be given in full or in part under the following headings
 - Employer pays contributions out of pre tax profit
 - Employer contributions not taxed as employee remuneration
 - Employee contributions out of pre tax income
 - Investment income not taxed
 - Investment capital growth not taxed
 - Pension not subject to tax
- Reduced contributions to the state retirement system
- State subsidies e.g. partial matching of contributions, per capita payments
- Guarantee to supplement investment returns if a predefined return is not achieved
- Make the private provision of benefits compulsory.
- Impose a maximum on the charges that can be imposed by investment / administration / product
- Providers.
- Make provision for a centralised contribution collection agency.
- Make provision for a centralised administration system.
- Make regulations simple.
- Improve education/publicity
- Reduce state benefits
- Legislate to improve security of private pension arrangements

(4)

ii)

- Impose restrictions on the size of funds which may be accumulated
- Restrict the types of investments on which tax relief may be given
- Limit tax concessions to benefits / contributions based a maximum level of income
- Require
 - Administration procedures to comply with minimum standards
 - periodic valuation / reporting
 - the arrangements to be supervised by someone independent from the employer / provider
 - disclosure of information to (potential) beneficiaries
 - benefits to be funded
 - assets to be kept separate from those of the employer
 - Professionals involved with the running of the scheme
 - to report wrongdoing / non compliance

(3)

iii)

Individual entry age method

For each age calculate the present value of benefits to be accrued over future service

Allowing for salary increases, investment income and, if appropriate, expenses of running the scheme.

Divide by the present value of 1% of a member's salary

(3)

iv) a)

- Salary progression
- Investment income
- Mortality of members post retirement
- Proportion married at death (after retirement)
- Mortality of widow(er)s
- Expenses
- Mortality of members before retirement and probability of leaving the scheme would be ignored
- as, otherwise, if the member stayed in the scheme till retirement the fund would be insufficient to provide full benefits.
- Inflation

b)

- Need to check what types of investment classes will be allowed
- Need to compare the correlation of investment return with general salary growth.
- May be able to get information about employee salary growth in the past but have to consider relevant for future.
- If there has been no private pension provision in the past there are unlikely to be any sensible mortality statistics for scheme members
- Population mortality may be available but need to consider if the employee mortality is likely to be different
- The statistics maybe out of date
- Would want to consider more than one set of mortality statistics to see if there are any trends and to consider if they should be allowed for in the future
- The timescale is very long so make allowance for changes in mortality would be sensible
- Need to consider company's attitude to risk

(6)

v)

- Contributions are paid over many years and accumulate after deduction of expenses at the rate of return earned by the fund.
- The contributions paid will be a percentage of salary from time to time.

- The desired benefit will depend on the level of salary at retirement.
- The cost of providing that benefit will depend on the investment return earned on the assets held
- the rate of inflation after the member retires the number of years the member survives in retirement
- whether the member is married at date of death how long the widow(er) survives thereafter the expenses of operating the assets and paying the benefits
- None of these factors can be known in advance
- The rates calculated are based on assumptions which may, on average, approximate to the actual experience but which are unlikely to be exact for any member.
- If the level of benefits is critical then, either contribution rates will have to be reviewed during the period up to retirement or the scheme will have to operate a reserve fund of some kind.
- If this is not acceptable then some members will retire on higher than desired benefits and some on lower.
- In any case, the fund will only be able to guarantee benefits once they have commenced if a suitably funded arrangement is set up.
- Targeting one benefit at retirement may under/over provide in other circumstances e.g. withdrawal, death, early retirement, late retirement
- There could be legislative changes in future which could impact on the investment return/benefit level

(6)

[22 Marks]

Solution 2:

i) Funding Valuation:

Usually assumes the scheme will be continuing

Primary purpose is to provide advice about the future level of contributions

There is no single correct way to ensure that a pension scheme will provide the promised benefits

The objectives of the valuation also include:

- Comparing assets & liabilities to assess the degree of security for the benefits
- To review the financial progress of the Scheme since the previous Valuation
- To determine an appropriate investment policy

It is common for the valuation assumptions to be relatively cautious

It is important that the valuation of assets & liabilities are consistent

Discontinuance Valuation

The purpose is to assess how secure the members' accrued benefits would be if the scheme had to be terminated at the valuation date

All active members will be treated as early leavers with benefits calculated with reference to salary at the valuation date (perhaps with some allowance for increases thereafter)

Benefits could be secured by purchase of insurance policies

Hence there is no real subjectivity in the calculations or the scheme could be run as a closed fund adopting an appropriate investment policy to minimise the investment risks

(6)

ii)

Discounted cashflow funding valuation

Assets are valued by estimating future income (dividends, rents etc) and discounting them to give a present day value

The discount rate used is consistent with the long term expected investment returns from the assets held

This is often derived by considering the real return on equities relative to the long term expected inflation rate and then making a deduction to reflect the other assets held

The liabilities are valued using the same long term rate of interest As a result:

- the valuation of the assets and liabilities is on a consistent basis as the same long term discount rate is being used
- and market fluctuations in asset values are smoothed

Market Value Approach

The assets are taken at market value

To ensure a consistent approach a market value of the liabilities is needed

There are a number of different approaches to determining a suitable discount rate

These include:

- Discounting liabilities at bond yields

- Or allowing for an equity premium to take account of potential higher returns from investments such as equities

(5)

iii)

Why are different methods used at different times

A market value approach makes more sense in the context of a discontinuance valuation because if the scheme were terminated at the valuation date the actuary would be testing to see if assets held at this point would be sufficient to buy the benefits

Accounting valuations often require assets to be taken at market values hence the overall valuation basis would be market based

The discounted cashflow approach produces a reasonably stable contribution rate over the long term

Whereas the contribution rate produced by looking at market conditions may not reflect the long term cost of the scheme

Hence a market value approach might also use a long term rate to determine the basic contribution to cover benefit accrual in the future or alternatively it might use a short term rate

Depending on the circumstance of the scheme it might be prudent to set a contribution rate for the short term that is higher than the long term average

(3)

[14 Marks]**Solution 3:**

i)

Lower yields lead to a higher cost of financing pension & other long term benefits for a given level of defined benefits for example a typical joint life annuity might cost around 30% more if expected long term inflation and hence nominal bond yields fall by 3%

There are two practical choices:

Pay more may be unpalatable to scheme sponsors

Invest in higher returning assets and hence riskier assets e.g. investing in corporate bonds rather than government bonds to target the added yield.

The promised benefits from completed service are generally clear and hence will not be affected

But scheme sponsors will review the additional cost and future benefits may be modified or the scheme replaced by a defined contribution arrangement

Schemes with fixed pension increases do not have any flexibility to adapt to periods of unfavourable experience

Benefit design with a low discretionary content is inflexible hence sponsors may consider increasing the discretionary content e.g. pension increases

When the fixed increases are greater than the inflation rate it is likely there will be rises in the real value of pensions

Similarly periods of deflation would see rises in the real value of pensions even if the scheme had no escalation of pensions in payment

For many members it could result in a higher real pension at retirement than might have been anticipated

For scheme that secure benefits at retirement by the purchase of an annuity, lower expected inflation and low real investment returns will be reflected immediately in more expensive annuity rates

(8)

ii)

Members will be directly affected by a long term change in inflation However, an analysis of true costs will usually be based on contributions, benefits and expenses which are fixed relative to inflation

The level of inflation is irrelevant and the expected costs or resulting benefits will only depend on the real returns & real level of expenses

In conclusion lower inflation has no meaningful effect on the value of the defined contribution arrangement

But real returns may differ according to the level of inflation

Many pensioners buy level pensions and even with very low inflation the pension will be severely eroded for long lived pensioners

With low inflation there is more chance that inflation will subsequently rise in the future

The propensity to save may differ in a future period of low inflation (logically or illogically)

The level of expenses charged might differ in a low inflation environment, although in real terms they should be similar

Different generations of members may achieve wildly different pensions for effectively identical contribution patterns

As mortality improves the cost of the pension increases. This increase for level pensions is more significant in times of low inflation than in high.

(6)

[14 Marks]

Solution 4:

i) Assumptions needed for a valuation of a defined benefit pension scheme might be:

Economic factors

- discount rate (which may be based on expected investment return)
- dividend yield
- dividend growth
- earnings inflation
- price inflation
- pension increases

Demographic factors

- rates of retirement in normal health
- rates of ill-health retirement
- rates of withdrawal (for reasons other than retirement or death)
- new entrant rates
- rates of mortality before and after retirement
- proportion married
- average age of dependants
- dependants' mortality
- promotional salary scale

(4)

ii) Different types of basis

Different actuaries use different terms for describing the actuarial basis – you may Encounter any of the following strengths of bases:

- a set of assumptions where each value is chosen as the median of the probability distribution is known as a best estimate basis (ie there is a 50:50 chance of the assumptions overestimating or underestimating future experience)
- an optimistic or weak basis is one that produces a high funding level. This may be as a result of producing a high value of assets and/or a low value of liabilities
- conversely, a cautious or conservative or strong basis produces a low funding level. This may be as a result of producing a low value of assets and/or a high value of liabilities
- a prudent basis usually lies in the range between a cautious basis and a best estimate basis.

When setting individual assumptions, an actuary must be careful to consider the effect of the assumptions as a whole. If each assumption is slightly on the cautious side of its best estimate value, the cumulative effect might be a basis that is very cautious overall.

(4)

iii) When using past data the actuary therefore needs to consider how to deal with:

- abnormal fluctuations due to demonetization etc
- changes of the experience with time
- Random fluctuations
- changes in the way in which the data was recorded
- Potential errors in the data
- changes in the balance of any homogeneous groups underlying the data
- Heterogeneity within the group to whom the assumptions are to relate.

Changes affecting economic data

- Economic data will fluctuate with changes in economic and fiscal policy.
- Past data for investment returns, salary levels and dividend yields in most countries will fluctuate significantly over an extended time-frame. It could be viewed that economic and fiscal changes mean that most past data is irrelevant (the effect of demonetization is a one off event and the lower interest rates may not be in the long term, as might have been perceived by the Actuary) and so only data that relates to periods after significant changes can be used. However, this would reduce the credibility of the data and increase the effect of any random fluctuation.
- Given the surplus in the scheme, the Actuary might have considered existing assets yielding higher returns in making the interest rate assumptions

- It is therefore necessary to try to use the earlier data and to try to strip from that data the fluctuations e.g. Demonetization effect (one off event) that relate to economic and fiscal conditions that differ from those that exist currently.

(6)

iv) Reasons why the premium may have increased

- There may have been an increase in scheme membership or in the salaries of members, leading to an increased premium in absolute terms although not in real terms.
- The membership of the scheme may be ageing, leading to a higher premium.
- The experience of schemes in general may have worsened over time, for example a new illness may have emerged leading to heavier mortality experience, and this is being reflected in the premium rates.
- The experience of this particular scheme may have worsened over time and this will be reflected in the premium rates if an experience rating approach is being used.
- There may have been a change in the insurance pricing structure or the provider may no longer be competitive.
- The increase in rates may reflect the position in the insurance cycle, ie the market may be less competitive than in the past and so insurers can incorporate larger margins for profit, expenses and contingencies in their pricing basis ...
-this may particularly be the case for the partner's annuity for which the market is less competitive.
- Insurance companies may have made losses in the past that they are now aiming to recoup.
- The premium for the partner's pension reflects:
 - expected investment return over the period for which the annuity would be paid, if future investment returns are expected to be poor then the premium will increase
 - the partner's life expectancy, mortality improvements may increase the cost of purchase.
- There may have been a recent benefit improvement, eg lump sum death-in-service benefit from four times salary to five times salary.
- There may have been changes to legislation or taxation being factored into pricing.

(3)

v) Possible ways of reducing the premium

The premium will be reduced if the benefits provided are reduced. This could be achieved via:

- a reduction in the multiple of salary provided on death (ten times salary is generous, perhaps reduce to five times salary)
- a reduction to the percentage of member's pension provided as a partner's pension (80% is very generous, perhaps reduce to 50%)

- a reduction to the number of years included in the calculation of the partner's pension – a prospective pension is expensive if a young member dies in service, perhaps limit the number of years used subject to age and/or service criteria. When considering a reduction in benefits the director must bear in mind the benefits offered by competitors and the likely impact on industrial relations of making such cuts. The eligibility conditions for receiving benefits could be tightened to reduce costs via:
- carrying out some initial medical underwriting on joining the scheme or becoming eligible to receive protection benefits, which may mean the insurer is prepared to offer better terms
- only providing full benefits for members who have completed a certain number of years' service.
- The premium may be reduced by choosing to operate experience rating (if this isn't already being used). This will be the case if the company believes their experience is better than that of the average company on which the premium rates are based.
- Premiums may be reduced if the company opts for a lower free cover limit so the insurer carries out additional underwriting to reduce risk.
- Premiums may be reduced by changing insurers, it may be that the insurer currently used is expensive relative to competitors.
- The impact of the cost of the higher premiums could be mitigated by requiring members to meet part of the cost, this may be as part of a flexible benefits arrangement where members pay more towards higher levels of protection benefits.

This is a very large scheme and so experience is likely to be fairly stable over time. Such a large scheme may wish to consider:

- Self-insuring all benefits so as not to pay towards profit and expense margins of the insurers. However, this needs to be weighed against the increased volatility of costs, liquidity concerns (particularly for the lump sum death benefit) and extra administration that arise if insurance is not used.
- Only insuring the lump sum death benefit (since this is the benefit that causes liquidity concerns and has the more competitive market).
- Only insuring certain members, ie those with significant liabilities, eg death-in-service lump sum for executives and young members (for whom the death benefits would be far in excess of the past service reserves held).
- A scheme that self-insures is at risk of a catastrophic event, eg an explosion at the company killing several employees, this risk can be reduced by taking out catastrophe cover.

(4)

[21 Marks]

Solution 5:**i) Options available at discontinuance of Scheme:****Options**

If a benefit scheme is being discontinued, the following options may exist for the provision of the outstanding benefit payments:

- gradual removal of liabilities by the continuation of the scheme without any further accrual of benefits but this may not be a good suggestion as the employer's covenant is weak.
- transfer of the liabilities to another pension scheme with the same sponsor is also not a good option given that the employer's covenant is weak and being a middle sized company it might not have another pension scheme.
- transfer of the funds to the beneficiary to extinguish the liability
- transfer of the funds to an insurance company to invest and provide a benefit
- transfer of the liabilities to an insurance company to guarantee the benefits
- transfer of the liabilities to a central discontinuance fund, operated on a national or perhaps industry-wide basis.

(3)**ii) Transferring funds to beneficiaries**

The issues for the trustees:

- Security of the scheme – the ability of the scheme to provide benefits. The impact on the security of the scheme for the remaining members of this, or will members receive reduced transfer values? Are more funds available if there is a deficit?
- Use of the funds – how will members use the funds? Will the funds be squandered, or will there be any restrictions on the use of the funds?
- Fairness – will this option result in all categories of members being treated equitably. Are there any winners or losers against other possible options?
- Scheme design – what allowance is made for complex and discretionary benefits under this option and the other options? Are material individual characteristics taken into account in some options? What is their worth?
- Investment policy – assets will need to be realised and this could result in uncertainty where there are volatile assets and potentially high dealing costs. Is it a good time to realise assets in the market?
- Risk – DB risks will be removed from the scheme and the risks will be transferred to the beneficiaries for those who take up this option.
- Advice – is there a need to provide advice to members as to whether to take this option and, if they do, subsequently what to do with the funds?
- Legislation, administration and communication considerations.

(4)

iii) Setting the terms

Regulation

- Regulation may dictate the approaches permitted on discontinuance.
- If this approach is permitted then there may be legislation that will affect the terms. For example, anti-discrimination legislation may mean that unisex factors must be used or the basis may be prescribed to some degree. In addition, there may be minimum and maximum amounts that can be paid.

Security

- The terms should be set so that the exercise of this option should not jeopardise the financial security of the scheme.
- Terms could be set using the ongoing funding basis so that the exercise of this option would have no effect on the ongoing funding level. Alternatively, a best estimate basis could be used so as to be cost neutral in terms of the actual cost of benefits.

Funding position

- The funding position of the scheme should be considered, if it is underfunded and the employer is not in a position to make up the shortfall then the funds to be transferred to the beneficiaries will be reduced.
- If the scheme is in surplus, then ownership of the surplus will need to be considered and, in particular, whether the funds to be transferred will be augmented.

Selection

- The terms for this option should be set to minimise any risk of selection, *eg* those in poor health could be more likely to take the fund if the terms do not reflect their health status.

Sponsor's view

- The sponsor's view needs to be considered. For example, the sponsor may wish to encourage members to take this option so that the DB risk is removed and therefore the terms could be set to be generous.

Fairness

- The terms should be set to provide value for money in respect of the DB benefits being given up. The terms should also be fair to members who take, and who do not take, this option. This suggests cost-neutral terms.
- Equity should be considered against the terms offered on any alternative options.

Market conditions

- The terms are likely to reflect current market conditions as liabilities are being extinguished and this is a short-term valuation. Thus the current investment policy will need to be considered.

Simplicity

- The terms should be simple so that the option is simple to administer and communicate. This may lead to terms which do not reflect individual circumstances; this could increase the risk of selection.

Expenses

- The trustees need to agree who will meet the expenses of offering this option, and therefore whether the terms need to include an expense allowance.

Take up rate

- If it likely that most members will take up this option, then selection is less of a risk. In addition, if assets need to be sold, the terms may need to reflect the cost of a quick bulk sale of assets.
- If few members are likely to take up this option then individual terms (to reflect the selection risk) or broad brush terms (to minimise the marginal expenses of the exercise) may be used.

Use of funds

- The terms may reflect how the funds would be used. For example, if members are required to purchase a deferred or immediate annuity with an insurance company then the terms may reflect the terms upon which such contracts are available.

Options

The trustees need to consider how to allow for options that are not cost-neutral, *eg* commutation factors. Should it be assumed that all/none take-up the options or allow for a proportion?

(8)

[15 Marks]

Solution 6:

i) The liabilities are extremely long term. The actual timing of the payments is unknown. Further the in-service liabilities increase as salaries increase and the pensions in payment are likely to also be increasing to some degree in line with inflation. The amounts of the future liabilities are therefore unknown. Some liabilities may crystallize on discretionary events — for example a highly paid, long serving member leaving service or electing to take early retirement.

The total contributions may be volatile. If members contributions are fixed then the company's contribution will be geared and thus more volatile. While the returns from fixed interest stocks are known, there may be a default risk. The returns from equity and property investment are unknown, and assumptions will need to be made.

(4)

ii) Because of the uncertainties in amount and timing of both asset proceeds and liability outgo, full matching of cash flows will not be possible. One reason is that there may not be assets available of sufficiently long duration.

Over the long term shares are generally considered to be the most appropriate asset to meet long-term real liabilities. Indexed linked bonds would also be appropriate. For fixed liabilities such as pensions in payment, a substantial proportion of fixed or index linked bonds would be appropriate.

Most asset types have uncertain returns. If the changes are not managed over time then there would be large trading costs associated with one-off changes. The costs of trading to adjust the portfolio frequently to achieve full matching may be prohibitive.

Given infinite resources, it is possible to try to meet outgoings by buying excessive amounts of securities.

In practice, the matching portfolio is the portfolio which costs the least and which still provides the required certainty of meeting the liabilities. The scheme may have to alter its investment guidelines to allow this.

(4)

iii) The current investment objective does not take any regard of the risk the investment manager may have taken. This may lead to volatile returns and lead to volatile contributions being needed.

The scheme may wish to reduce the relative risk of the investment strategy probably at the expense of some future returns.

The objective may not be a fair comparison if other schemes are required to operate under more restrictive conditions. For example other schemes may be required to match pensioner liabilities with fixed interest stock, even if full cash flow matching is not required.

The definition of “comparable benefit schemes” is not precise and is potentially easy to manipulate to achieve a top quartile return.

(3)

iv) The scheme could establish an asset allocation benchmark that reflects the proportion of pensioner liabilities to liabilities relating to current employees.

The part of the benchmark relating to fixed liabilities could have a high proportion of fixed interest and index linked securities. The portion relating to “real” liabilities could be heavily weighted to equities and property.

Investment could be restricted to companies with a minimum credit rating.

High risk investments such as speculative use of derivatives could be prohibited.

Requiring a minimum number of individual holdings in certain categories (e.g. domestic equities) would increase diversification and reduce risk, but might require the manager to hold stock that he did not consider appropriate, merely to have sufficient lines.

The scheme could require that the market value of individual holdings remains within a pre-agreed range. As market values change and these ranges are exceeded steps would be taken to rebalance the portfolio within a reasonable time frame given costs and market opportunities.

Further, if the scheme's liabilities are solely domestic then it may be reasonable to constrain the fund to limit investment in foreign shares and fixed interest investments. The benefits of further diversification may not outweigh the currency risks. This constraint would be expected to reduce both the risk and returns from the fund.

(3)

[14 Marks]
